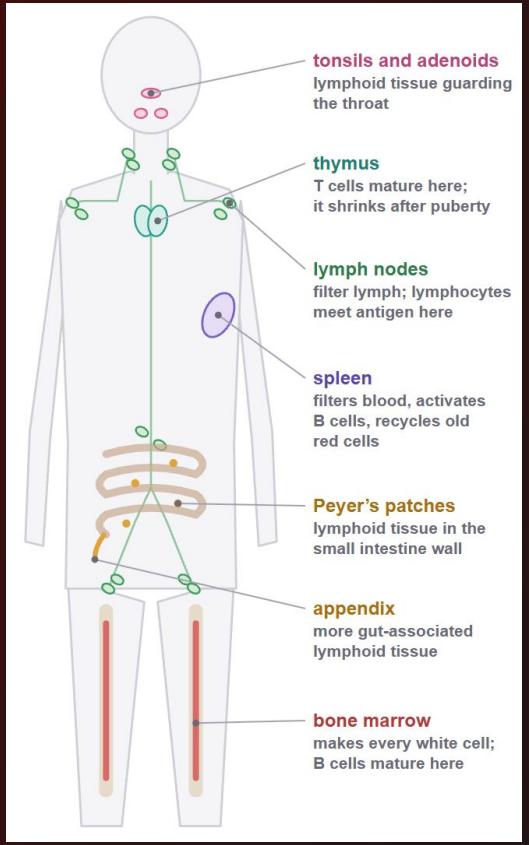
The Immune System

M-CATS at UK

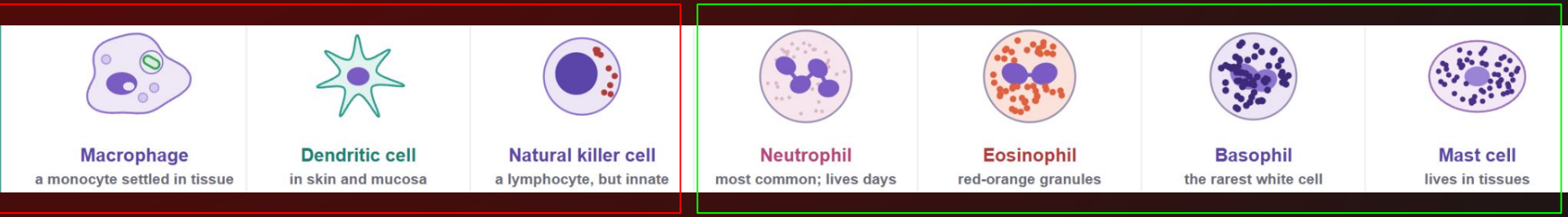
The Structure of the Immune system

- Components
  - Tonsils
  - Adenoids
  - Spleen
  - Peyer's Patches
  - GALTs
  - Bone Marrow
  - Thymus
  - Lymph Nodes



Cell Types

- Leukocytes are derived from hematopoiesis, all from hematopoietic stem cells
  - Granulocytes vs Agranulocytes
- Monocytes ↠ Macrophages (e.g. microglia, osteoclasts)
- B cells and T cells



Divisions of the Immune System

- Innate
  - Nonspecific, rapid response
- Adaptive
  - Specific, slower initially, but more effective
    - Humoral and Cell Mediated

A closer look at B and T Cells

B Cells

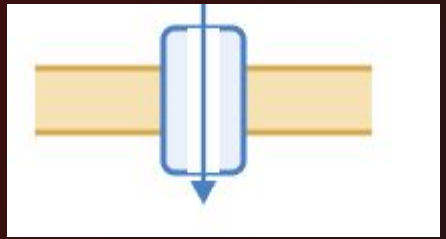
- Mature in the Bone Marrow
- Activated in the spleen or Lymph tissues
- Turn into plasma cells that secrete antibodies
- Humoral immunity

T Cells

- Mature in the Thymus
- Activated through MHC interactions… We'll talk about this later
- Cell mediated immunity

Innate immunity

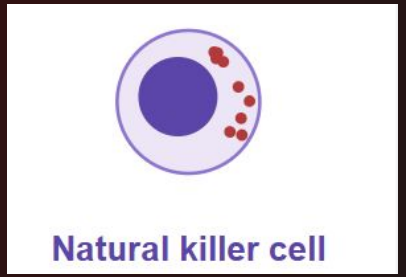
- Skin/Mucous membranes
  - Lysozyme
  - Defensins
  - Sweat
  - Mucus
- GI system
  - Native flora
  - Acidic environment
- Complement System
  - ‘Hole puncher’
  - Bacterial defense system
  - Two activation methods (antibody dependent vs not)
- Interferons
  - Viral defense system



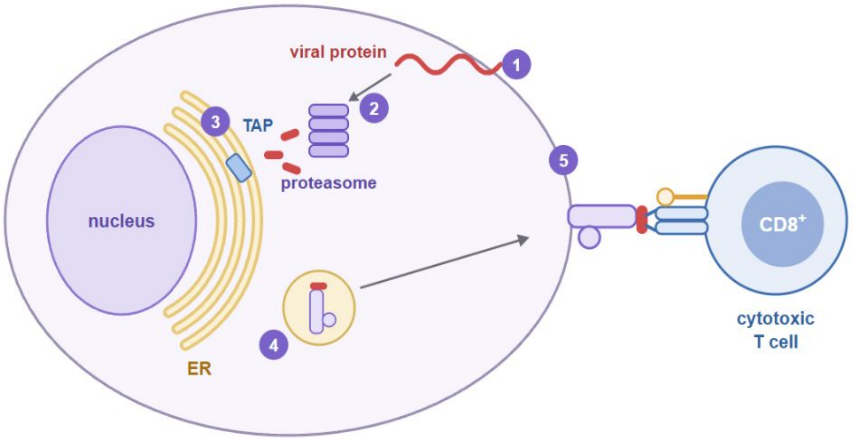
Cell Types of the Innate System



- Macrophages
  - Permanent residents of specific tissues
  - Phagocytizes and digests invaders
  - Uses **MHC-II** proteins to show the adaptive immune system proteins from the invader
  - Releases cytokines and inflammatory factors to recruit other immune cells to the area
  - Along with dendrites, has Pattern Recognition Receptors (PRRs) which classify the type of invader
- Natural Killer (NK) cells
  - Flawed MHC detection systems to kill cancerous or infected cells



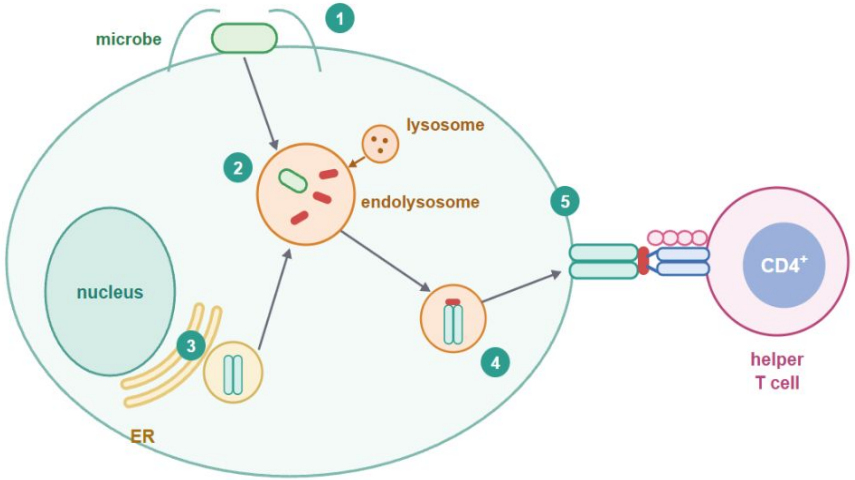
MHC class I: what a cell is making



1. Any protein made in the cytoplasm, including a virus's, is sampled.
2. A proteasome chops some of it into short peptides.
3. A pump (TAP) moves the peptides into the ER.
4. Each peptide is loaded into the groove of a new MHC-I molecule.
5. The pair goes to the surface, where a $CD8^+$ T cell checks it.

A foreign peptide marks the cell for killing; a healthy cell shows only self peptides.

MHC class II: what a cell has taken in



1. A dendritic cell, macrophage or B cell takes in a microbe.
2. Lysosome enzymes cut it into peptides in an endolysosome.
3. New MHC-II molecules arrive from the ER and Golgi.
4. A peptide is loaded into the MHC-II groove.
5. MHC-II shows it at the surface to a $CD4^+$ helper T cell.

A helper T cell that fits releases cytokines that switch on B cells, killer T cells and macrophages.

Cell Types of the Innate System Cont.

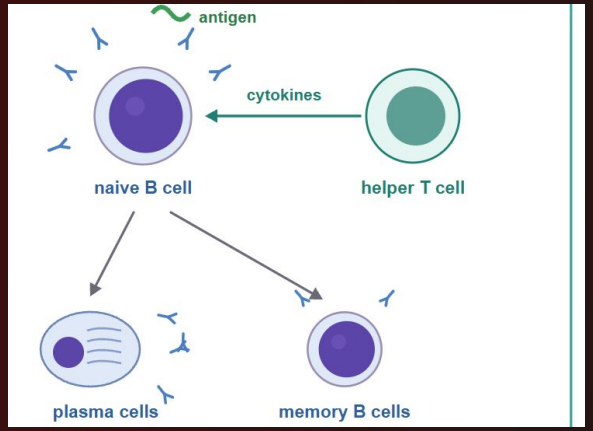


- Granulocytes
  - Neutrophils
    - Most populous but short-lived
    - Use chemotaxis to detect bacteria
    - Recognizes opsonized bacteria
  - Eosinophils
    - Mediate allergic reactions and parasitic infections
    - Release histamine, resulting in ‘leaky’ vessels and vasodilation (why?)
  - Basophils and Mast cells
    - Mainly involved in allergic reactions
    - Closely related, but mast cells exist primarily in the tissue, mucosa and epithelium
    - Both release inflammatory factors during a reaction



The Adaptive System

- B and T cells
  - Various different types with related, but distinct functions
  - The adaptive system relies entirely on these cells, so several diseases targe this system
  - ‘Learning’ system



The T-cell family

Helper T cells
CD4+

Read MHC-II. Release cytokines that switch on B cells, cytotoxic T cells and macrophages. Th1 cells mostly arm macrophages (interferon-γ); Th2 cells mostly help B cells, especially against parasites.

Cytotoxic T cells
CD8+

Read MHC-I. Bind an infected or cancerous cell and release perforin, which opens pores, and granzymes, which trigger apoptosis, so the cell dies without spilling its contents.

Regulatory T cells
CD4+

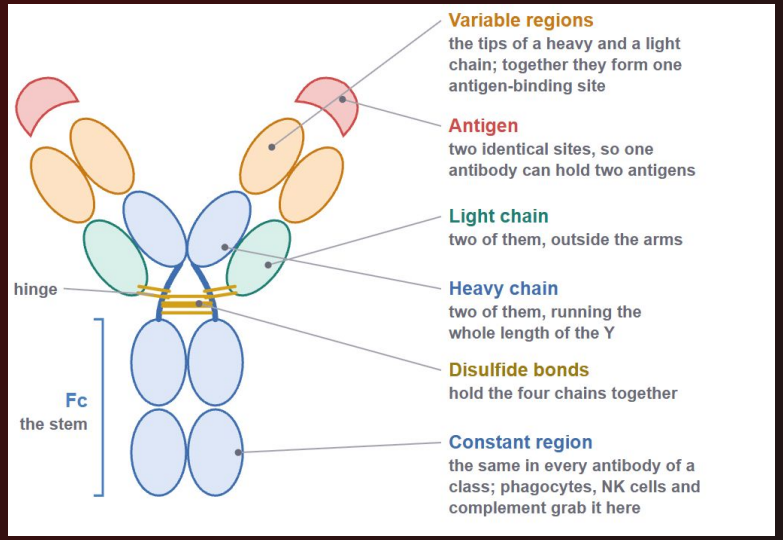
Also CD4+, marked by Foxp3. Calm a response once an infection is controlled and hold back self-reactive cells that escaped the thymus, preventing autoimmunity.

Memory T cells
CD4+ or CD8+

Long-lived cells left after an infection. On a second encounter they respond faster and harder, like memory B cells.

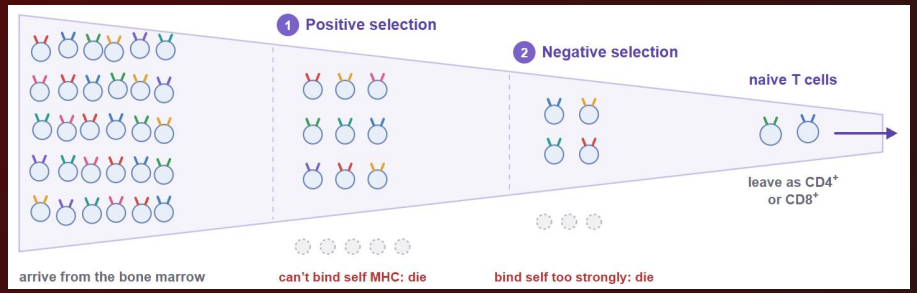
B Cells

- B cells originate in the bone marrow and start out mature, but naive
- They are responsible for secreting antibodies (Ig) that bind to antigens
  - Antibodies serve 3 functions
    - Opsonization
    - Agglutination
    - Cell Surface binding
  - 5 isotypes (IgA, IgM, IgD, IgE, IgG)
    - Cells can switch between isotypes based on molecular signals
- After exposure to an antigen, B cells undergo hypermutation to create an optimized antibody
- Plasma cells and Memory Cells



T Cells and cytotoxic immunity

- T Cells mature in the thymus and undergoes two checks
  - Positive selection-ensuring the cell recognizes MHC complexes
  - Negative selection-ensuring the cell doesn't target self 'antigens' on MHC complexes
- T cell maturation is facilitated by thymosin, and when mature, T cells initially are also naive, undergoing activation when exposed to antigens

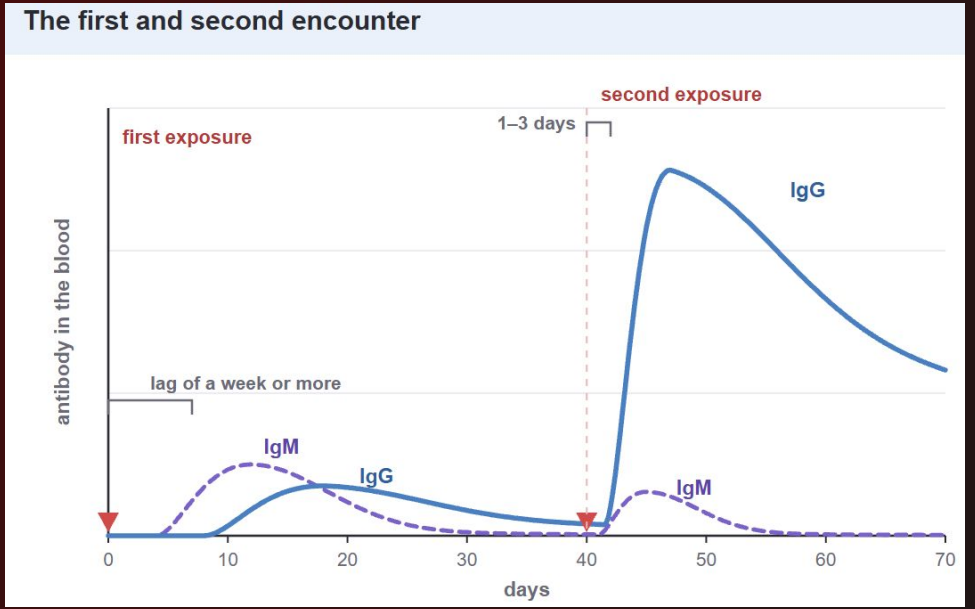


T Cells Cont.

- Helper T cells ($CD4^+$)
  - Respond to MHC II antigens by recruiting immune cells via Lymphokines
  - Activates B cells
  - Targeted by HIV
- Cytotoxic T cells ($CD8^+$)
  - Respond to MHC I antigens and kill cells that present foreign antigens
- Regulatory T cells
  - Shuts off immune response machinery once infection is contained
  - Turns off self-reactive lymphocytes
- Memory T cells
  - Functions similarly to B memory cells, storing info about past invader

Primary Vs. Secondary Response

- Memory cells help the body mount a stronger response upon re-exposure
- Most of the lag is due to B and T cell hypermutation to create optimized antibodies and responses
- Vaccines ensure a secondary response by simulating a primary exposure without actual infection



Tracing the path of viral vs extracellular bacterial infection

Innate immune system reaction?

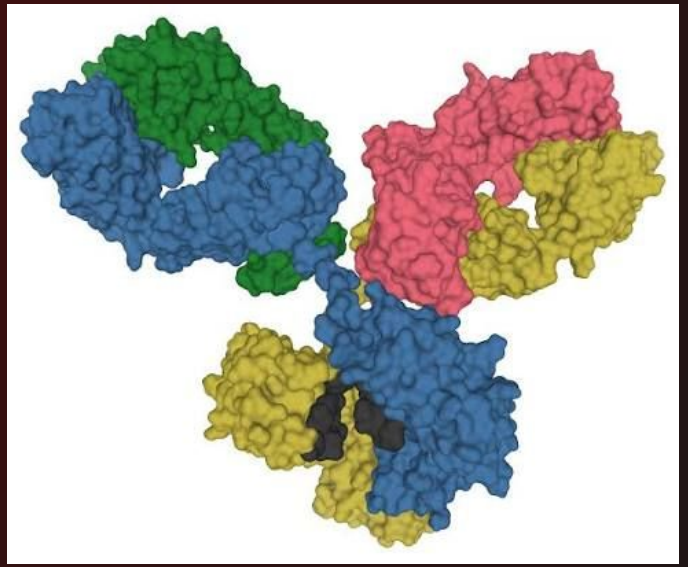
Adaptive system reaction?

What kind of T cell is going to be most active?

Which cells would mediate the initial response?

Immunization

- Active vs Passive
  - Whether the body produces antibodies itself (vaccines or infection) or antibodies are passed from an external source (transfusion or placental transfer)
- Natural vs Artificial



Lymphatic system

- Made up of directional vessels that collect lymphatic fluid and deposit it into a vein close to the heart
- Nodes are locations where these vessels meet with arteries and veins to allow immune cells to interact with each other. B cells proliferate and mature in the lymphatic system
- The system also acts as a way for chylomicrons to be transported
    - Lacteals (small vessels) in villi then connect to the lymphatic system, allowing systemic transport
- The lymph system allows for fluid equalization
    - Blood to tissue fluid distribution is based on both oncotic and hydrostatic forces. Due to a slight mismatch in these forces, some fluid can remain in tissue. Also in the case of disease, a drop in oncotic force can be compensated for

Introduction

In the 1960s, lymphocytes were first parsed into two subsets, B cells and T cells. This was not without controversy; it was initially noted publicly that B and T were the first and last letters of 'bullshit'[1]. The 1970s saw the separation of T cells into two flavours — $CD4^+$ T cells and $CD8^+$ T cells — based on their expression of either the CD4 or CD8 co-receptor for the T cell receptor (TCR) (refs. [2,3,4]). All of these populations could rightly be defined as distinct cell types, as one does not typically see interconversion under physiologic circumstances. T cells were also segregated into naive, recently activated or effector, and memory T cells based on antigen-experience, which had a clear conceptual logic.